Clash of Clans Base Archetype Classifier

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Colab Notebook: `https://tinyurl.com/4928ku8s`
GitHub Link: `https://github.com/SattiDaBeaver/APS360-Project`

Project Description

This project presents a "Clash of Clans Base Archetype Classifier", utilizing Convolutional Neural Networks (CNN) and Vision Transformers (ViT) to identify the archetype of the base from an image of the base layout. The project aims to assist attack strategy analysis and planning in both casual and competitive Clash of Clans (CoC) battles. A base consists of walls, defensive structures, and other buildings arranged in geometric layouts. An example of a CoC base is shown in Figure 1a.

In Clash of Clans, a game with a large monthly player base of 80 million players, specific attack strategies are stronger against certain base archetypes, such as box, ring, compartment, and island. Accurately identifying the base archetype provides a strategic edge to the attacker and can be very valuable, especially in competitive CoC matches. However, identifying the correct archetype can be non-trivial due to the visual complexity and limited time in preparing for attacks.

Deep learning is a suitable approach for this task as the archetype of the base is primarily defined by the geometric layout of the walls, rather than individual placement of other buildings. CNNs are good at identifying local features like wall patterns and compartments, while ViTs are good at identifying global features like the overall base structure. This combination allows the model to distinguish visually similar bases that differ in wall structure, making deep learning effective for this task.

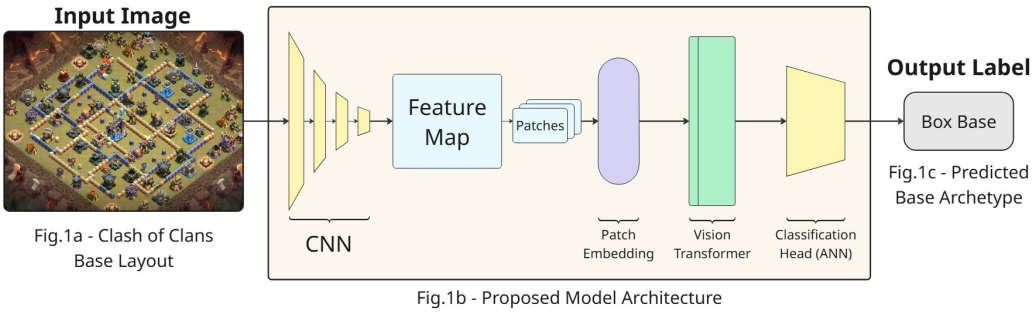

Figure 1: Proposed hybrid CNN-ViT model architecture

Data Processing

Data Collection

Images were collected from two public Clash of Clans base sharing websites: Clash Champs [1] and CoC Layouts [2], using a custom Python web scraper run locally. The script uses Selenium and BeautifulSoup for image parsing, and curl for downloading images. Using curl bypasses the Cloudflare bot protection. All images were scraped for Town Hall Level 18. Most images from Clash Champs are 1920×872px (varying slightly by base), while images from CoC Layouts are 1000×460px. In total around 2500 images were collected from the websites. Additional data will be collected from different town hall levels (like town hall 17).

Data Cleaning and Labeling

Upon inspection, most of the images collected from Clash Champs were mislabeled. Due to this, the images were manually sorted. Ambiguous and garbage images were discarded while labeling. Currently, 616 images have been labeled, and more will be labeled in parallel with model development. The class distribution is shown in Table 1. The four archetypes are: **Box** (1 or 2 large walled sections); **Ring** (light center with thick outer wall); **Compartment** (interconnected maze of small walled sections); and **Island** (multiple isolated walled pockets).

	Box	Compartment	Island	Ring
Train	82	226	87	97
Validation + Test	21	57	22	24

Table 1: Class distribution (80/10/10 split)

Data Processing and Augmentation

All data processing, model training, and evaluation was performed in Google Colab. Each image was resized to 300px on the shorter side, center cropped to 256×350px to remove UI banners and borders, then resized to 224×224px for ResNet compatibility. This pipeline also converts the isometric base view to a roughly square layout. During training, the images are randomly horizontally flipped for augmentation, effectively doubling the initial dataset size. To address class imbalance, a weighted random sampler was used. An example processed image is shown in Figure 2.

Figure 2: Left: Raw scraped image; Right: Processed image after resizing and cropping

Challenges

The primary challenges faced during data processing were:

1. Cloudflare bot protection on Clash Champs blocked the scraping scripts, requiring curl to bypass it.
2. Base labels were inconsistent/non-existent, requiring manual labeling of images (time consuming).
3. Class imbalance, with compartment bases being overrepresented in online datasets.

Plan for Unseen Data

10% of the images are reserved as the test dataset, and will be used to evaluate the final model. Additional unseen data will come from personal in-game screenshots or images of other town hall levels from Clash Champs, which were not used for training.

Baseline Model

A simple Artificial Neural Network (ANN) was used as the baseline model. The processed images are flattened to a 1D tensor of size $224 \times 224 \times 3$ (150,528), and passed through four fully connected layers of size 512, 256, 64, and 4, with ReLU activations. The output is a 4-class vector trained using cross-entropy loss. The model was trained for 30 epochs with batch size 32, learning rate 0.001, and Adam optimizer. The total number of parameters is ∼77.2M. The baseline architecture is shown in Figure 3.

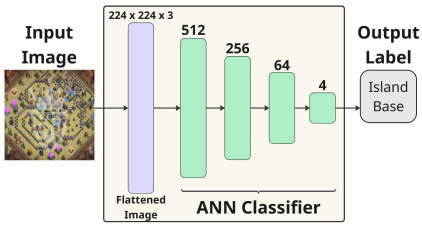

Figure 3: ANN baseline architecture

The ANN achieved a best validation accuracy of 47.5% (0.525 error), compared to a random chance of 25%. Due to the small labeled dataset, the model overfits easily, as evident from the rise in validation loss. The training and validation loss/error curves are shown in Figure 4.

From a qualitative standpoint, the ANN cannot capture the geometric and spatial features, lacking the local feature capturing of CNNs and the global pattern recognition of ViTs. The baseline is fast to train ($\sim$13 minutes for 30 epochs), providing a quick reference point for the primary model.

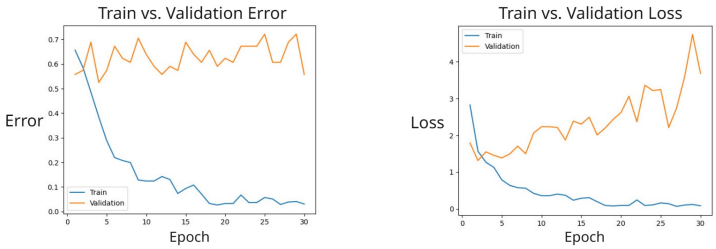

Figure 4: ANN training and validation loss/error curves over 30 epochs

Primary Model

The proposed final architecture is a hybrid CNN-ViT model. A pretrained ResNet CNN extracts local geometric features (like wall compartments), which are then passed to a Vision Transformer (ViT) to extract global features (like base structure). The output is then passed to a fully connected classification head for base archetype classification. The final architecture is shown in Figure 1.

The preliminary model omits the ViT and implements only the frozen ResNet18 CNN backbone with a classifier head. This model acts as a middle ground to test the feasibility of the proposed model against the baseline, and as a baseline for the final model (with the ViT). The 512-dimensional feature vector from ResNet18 is passed through: Linear(512, 256) $\rightarrow$ ReLU $\rightarrow$ Dropout(0.4) $\rightarrow$ Linear(256, 4). The frozen ResNet18 backbone has $\sim$11.7M parameters, while the trained classifier head has $\sim$132K parameters. The preliminary architecture is shown in Figure 5.

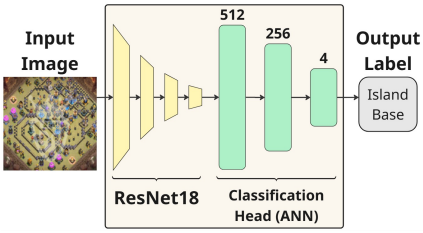

Figure 5: Preliminary model architecture

The model was trained for 30 epochs with a batch size of 32, learning rate of 0.001, and Adam optimizer. The best validation accuracy was **57.4%** (0.426 error) compared to the ANN baseline of 47.5%, demonstrating that CNN feature extraction captures spatial features that a flat ANN cannot. The model accuracy is still low and overfitting occurs quickly, likely due to the small labeled dataset. Qualitatively, the model identifies ring bases most reliably due to their visually distinct wall structure, while box and island are frequently confused with compartment. The training and validation loss/error curves are shown in Figure 6.

These are preliminary results. Future improvements include labeling more images to reach 300+ per class, fine-tuning 1-2 ResNet layers, and integrating the ViT for global feature recognition.

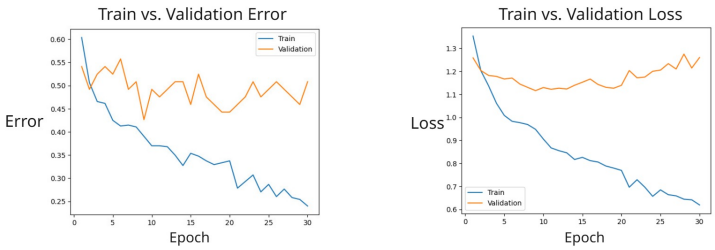

Figure 6: ResNet18 training and validation loss/error curves over 30 epochs

References

[1] Clash champs base database. https://clashchamps.com.

[2] Clash of clans layouts database. https://clashofclans-layouts.com.